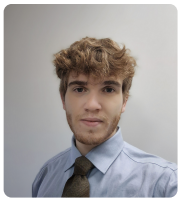
LÉHO DEHOVE

R&D ENGINEERING INTERN

BIO-MECHATRONICS AND ROBOTICS SPECIALIZATION

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Profile

Final-year Bio-Mechatronics Engineering student with hands-on experience in prototyping, machine learning, CAD, additive manufacturing, and electronics applied to robotics. Combines control theory, kinematics, and low-level firmware with neuroscience research (CNRS / ESPCI PSL) and industrial manufacturing (Danfoss). Seeking a 6-month R&D internship from April 2027.

Skills

Robotics, Vision & Control
ROS / ROS2, OpenCV, Linux
Orocos / BFL, MATLAB/Simulink
Kinematics, Dynamics, SLAM

Embedded & Electronics
C, C++, Python, Rust, Git
STM32, PIC16F1619, ESP32, Arduino
MPLAB X / MCC Melody, I2C/SPI, PCB

Prototyping & CAD/CAE
SolidWorks, CATIA, Abaqus/CAE, Rhino
3D Printing: 10+ years (FDM/SLA/SLS)
CNC Milling, Laser, Wood/Metal

Engineering Experience

CNRS - ESPCI PSL University (Brain Plasticity Lab) — June - expected Sept. 2026
Computer Vision R&D Intern

- Developing video processing software for automated kinematic analysis in animal behavior research, increasing throughput by 500%, saving over 45 hours per batch, and turning a week-long task into a single day.
- Co-authoring a research article on automated neural locomotion kinematics under optical stimuli.
- Designed and upgraded an optoelectronic hardware module for phototactic recording behavior.

Danfoss Power Solutions II (Manufacturing Plant) — June - July 2025
Manufacturing & Quality Operator

- Packaging, quality inspection, and assembly of small and medium-sized mechanical components for fluid power systems.

Enedis Business France (Energy Management Division) — July - Aug. 2023
Operations & Project Management Intern

- Supported national rollout of Linky smart meters through cross-team coordination, structured project tracking, and operational reporting.

ESME FabLab & Robotics Club — Sept. 2022 - June 2025
FabLab Administrator & Club Vice-President

- Managed digital fabrication machines (CNC, SLA/FDM 3D printing, laser cutters) and trained 150+ engineering students in CAD/CAM and rapid prototyping.
- Coordinated student club activities, cross-functional engineering projects, and robotics events.

Projects

3D Particle Manipulation Platform | **Custom Power Electronics** · **ROS** · **Microcontrollers** — 2025 - 2026

- Custom power electronics and ROS to control dust particles in a 3D enclosed space. Programmed deterministic control firmware.

Hand Exoskeleton for Rehabilitation | **Biomechatronics** · **Motorized Actuation** · **Embedded Control** — 2025

- Motorized hand exoskeleton to assist with daily muscle weakness and physical rehabilitation (functional physical prototype).

Autonomous Ball-Tracking & Mobile Robot | **ROS** · **OpenCV** · **Autonomous Navigation** — 2024 - 2025

- Autonomous robot capable of navigating, detecting, catching, and throwing a ball. Winner of the robotics tournament.

Upper Limb Prosthetic Device | **Biomechanics** · **Ergonomics** · **Material Physics** · **Additive Mfg** — 2023 - 2024

- Custom mechanical prosthetic and orthotic device developed for a real child with upper limb agenesis, focusing on ergonomics, material physics, and biomechanical fit without electronics.

Education

Master of Engineering - Bio-Mechatronics Specialization — Expected Sept. 2027
ESME Engineering School (Mechanical, Electrical & Electronics Curriculum)

Ongoing Key Courses : Medical Robotics, Biomechanics, Ergonomics for Design, Modeling and Simulation, Impedance Control, Robust Control, Biomaterials, Computer Vision, Real-Time Computing.

Exchange Student - Electrical & Electronic Engineering — 2025 - 2026
Universiti Putra Malaysia (UPM)

IEEE final year collaboration across international cultures | English Proficiency: C1 Academic English (IELTS 6.5 / TOEIC 875).

Additional Information

- **Languages:** French (native), English C1 (IELTS 6.5 / TOEIC 875), Spanish B1.
- **Interests:** Music (piano & guitar), Sports (weightlifting, climbing, seasonal), Video & board games (strategy), Raspberry Pi side projects, Science communication, 10+ years additive manufacturing.
- **Licenses:** Driving License (Category B and AM), Coastal Boating License (Permis Côtier).